

Measuring the footprint of large language models in the astronomy literature

SERAT MAHMUD SAAD¹

¹*Department of Astronomy, The Ohio State University, Columbus, OH 43210, USA*

ABSTRACT

Large language models became easy to use in late 2022, and their use in scientific writing has grown quickly in most fields. The size of this change in astronomy has not been reported. We measure it from the text itself. Using 200,547 astro-ph abstracts posted to arXiv between 2015 and 2026, we track a set of “marker” words that were rare for the whole decade before ChatGPT and then became common. The fraction of abstracts that contain at least one such word rose from about 1.5% before 2022 to about 5.8% in 2025, while a matched set of neutral astronomy words stayed flat. A conservative estimate puts a lower bound of about 4% on the fraction of recent abstracts whose language a model touched. We let the data pick the words and phrases without an imported list, and recover the same style vocabulary next to the genuinely new instrument and survey names. The marker words also appear together far more often than chance (mean pairwise lift 14, up to several hundred for some pairs), which is what one expects if a single cause acts on the whole abstract. Against these numbers, explicit disclosure of model use in astronomy papers stays below 1%, and crude checks of the LaTeX source find near-zero pasted-model text. We find no fabricated citations in a sample of recent papers, in contrast with a rising rate in biomedicine; the astronomy habit of building reference lists from the NASA Astrophysics Data System appears to protect the citation record. Finally, the word signal weakens in 2026 even as disclosure keeps rising, so word counts are a lower bound that will get harder to measure. All data and code are public.

Keywords: Astronomy data analysis — Scholarly communication — Natural language processing

1. INTRODUCTION

The public release of ChatGPT in November 2022 made large language models easy to use, and their use in scientific writing followed quickly. Because per-document detectors are not reliable, recent work measures the effect at the level of a whole corpus, by tracking words whose frequency changed abruptly after 2022. [Liang et al. \(2024\)](#) fit a statistical model to arXiv, bioRxiv, and the Nature journals and report that up to about 17.5% of computer science abstracts by early 2024 carried model-edited text. [Kobak et al. \(2025\)](#) count “excess” vocabulary in more than 15 million PubMed abstracts and find that at least 13.5% of 2024 abstracts were processed by a model. [Gray \(2025\)](#) estimate that a model was involved in more than 10% of all 2024 papers and describe a cat-and-mouse effect in which words lose their signal once they become known. Physics and mathematics sit lower in these studies, near 6–10%.

Astronomy has been studied less. [Astarita et al. \(2024\)](#) count model-favored words across about one million records in the NASA Astrophysics Data System ([Kurtz et al. 2000](#)) and show a clear rise after 2023, larger in non-refereed work; they do not convert the signal into a fraction. [Geng & Trotta \(2025\)](#) report the cat-and-mouse effect across all of arXiv. A survey of astronomers ([Fouesneau et al. 2024](#)) finds that most use these tools every week but few state so in print. No paper has reported, for astronomy specifically, how large the footprint is, how it compares with other fields, or how the different ways of detecting it agree.

The size of the effect matters for practical reasons. If a large share of abstracts are edited by a model, then word-based screening of papers or proposals will flag honest authors, and in particular non-native English speakers, at high rates. If model use is common but rarely stated, a reader cannot tell how a given text was produced. And if models draft text rather than only polish it, the record may drift toward a narrower and more uniform style. It is also useful to separate three uses that are easy to run together: a model as a copy editor, a model that drafts content, and a paper produced whole. The methods here mostly measure the first, and a measurement of its size is a first step toward any of these questions.

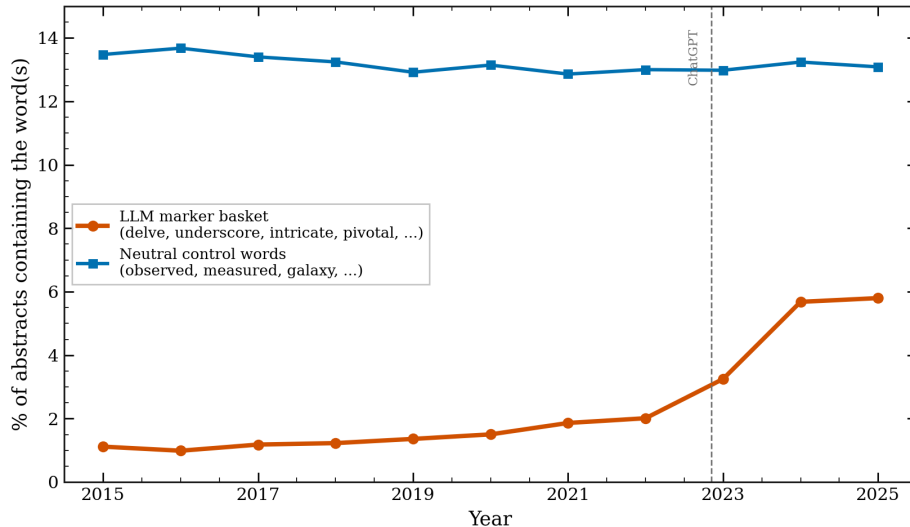


Figure 1. Fraction of astro-ph abstracts containing a marker word (red) and a matched set of neutral control words (blue), by year. The marker basket rises after ChatGPT while the control stays flat. Both are per-abstract fractions, so we stop at the last full year, 2025.

We provide those numbers. We harvest every astro-ph abstract from 2015 to 2026 and measure the marker-word signal, its rise, and its recent weakening. We convert the signal into a lower bound on the fraction of abstracts a model touched. We let the data select the words and two-word phrases without an imported list, and we measure how often the marker words occur together. We then compare the word signal against three other lenses on the same corpus: explicit disclosure in acknowledgments, pasted-model text in the LaTeX source, and fabricated citations. We close with cuts by subfield and by author country, a measurement of promotional versus hedging language, and the limits of the method. The point is narrow and worth stating up front: a marker word is a sign of model-edited language, not of poor or dishonest science.

2. DATA

We queried the arXiv API for every paper with a primary or cross-listed astro-ph category, month by month, from January 2015 through July 2026, and kept the version-1 submission date, the primary category, and the abstract. After removing duplicate versions this gives 200,547 abstracts. We tokenize each abstract into lowercase alphabetic words. A word’s “document frequency” in a year is the fraction of that year’s abstracts that contain the word at least once, which is the quantity used in earlier work.

For disclosure (Section 7) and for the by-country cut (Section 10) we use the NASA Astrophysics Data System, whose `ack` field searches the acknowledgments section and whose `full` field searches full text where available. We count `database:astronomy` records per year as the denominator.

3. THE MARKER-WORD RISE

We assembled a basket of marker words from earlier studies: `delve`, `underscore`, `intricate`, `pivotal`, `showcasing`, `meticulous`, and about thirty more. For a control we used neutral astronomy words such as `observed`, `measured`, and `galaxy`, which should not depend on writing tools. Figure 1 shows the two curves. The marker basket sat near 1.5% for the whole pre-ChatGPT decade and rose to about 5.8% by 2025. The control words held flat near 13% across the same years. The rise is specific to the model-favored vocabulary and is not a general change in how abstracts are written.

The aggregate rise hides a moving target. Figure 2 tracks individual words by quarter. The word `delve` rose to a peak in early 2024 and then fell back toward its starting level once it became a widely known sign of model text, while less famous words such as `underscore` and `leveraging` kept rising. The total marker rate did not fall with `delve`; the load shifted to other words. Naming a marker removes it, but the underlying habit stays, which matches the cross-field result of Geng & Trotta (2025).

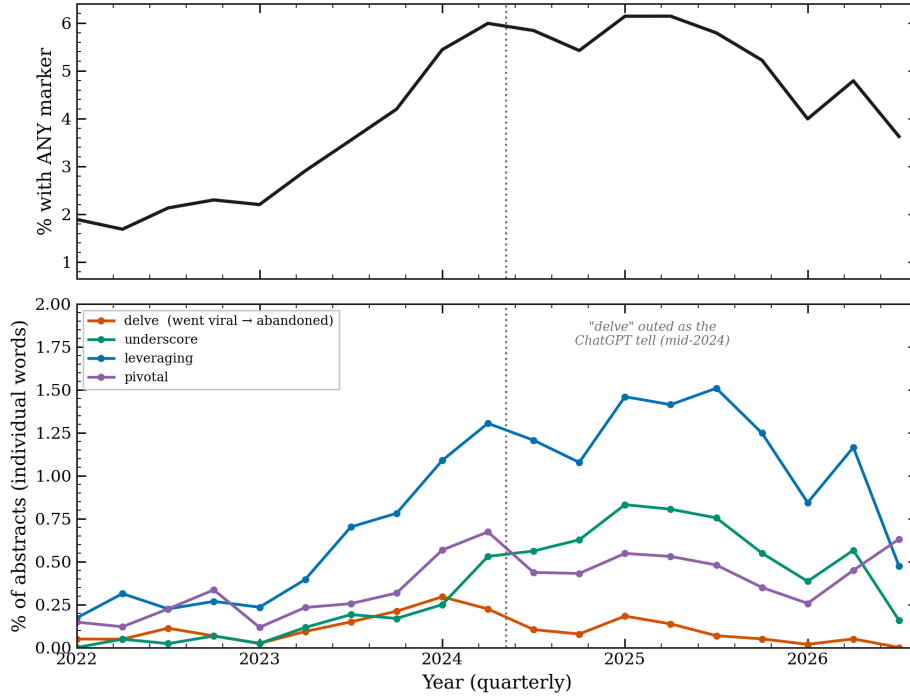


Figure 2. Top: the fraction of abstracts with any marker word, by quarter, keeps rising. Bottom: single words. **delve** peaks in early 2024 and then declines, while other markers keep rising; the aggregate does not fall with it.

4. LETTING THE DATA PICK THE WORDS

An imported word list invites the question of whether astronomy differs. We set the list aside and ranked every word by how much more common it became in 2024–2026 than in 2018–2021. The top of the ranking splits cleanly into two families (Figure 3). One family is the names of instruments and surveys that came online in these years (JWST’s NIRCam, MIRI, and NIRSpec; the DESI survey; XRISM), which is what a working method should find. The other family is the same style vocabulary reported in computer science and biomedicine, now recovered from astronomy on its own: **leveraging**, **pivotal**, **intricate**, **highlighting**. The imported list was not needed; the astronomy text points at the same words. Table 1 lists a representative set with their document frequency before and after ChatGPT.

The same split holds for two-word phrases. Figure 4 tracks a set of model-favored bigrams over time; **underscores the** and **highlighting the** rose by an order of magnitude, while a neutral control phrase (**wide range**) stayed flat. Ranking all bigrams by their increase again separates topic phrases (**jwst observations**, **the desi**) from style phrases (**offering a**, **results highlight**, **findings indicate**), as shown in Figure 5. Phrases carry a stronger signal than single words, and they are harder to remove by editing one word at a time.

5. THE MARKER WORDS TRAVEL TOGETHER

If the marker words shared a common cause, they should occur together more often than chance. For a pair of words a and b we measured the lift

$$\text{lift}(a, b) = \frac{P(a \cap b)}{P(a)P(b)}, \quad (1)$$

where $P(a)$ is the fraction of 2024–2026 abstracts that contain word a and $P(a \cap b)$ the fraction that contain both. A lift of one means the two words are independent. The mean pairwise lift over the marker set is 14, and about half of all pairs exceed one (Figure 6). Some pairs are extreme: an abstract that contains **delve** is a few hundred times more likely than chance to also contain **garner** or **tapestry**. Independent word drift would not produce this. A model that edits or drafts a whole abstract, and adds several of its preferred words at once, does. The clustering is direct support for reading the marker signal as a single effect rather than a set of unrelated trends.

Table 1. Representative marker words: document frequency in the pre-ChatGPT baseline (2018–2021) and the recent window (2024–2026), and the ratio of the two.

word	2018–21 (%)	2024–26 (%)	ratio
underscore	0.03	0.58	23
showcasing	0.01	0.12	10
leveraging	0.13	1.22	9
offering	0.26	2.08	8
delve	0.02	0.11	7
pivotal	0.08	0.47	6
highlighting	0.34	1.68	5
intricate	0.07	0.39	5

NOTE—Document frequency is the fraction of abstracts that contain the word at least once. The full table of every word that clears a count floor is in the public data release.

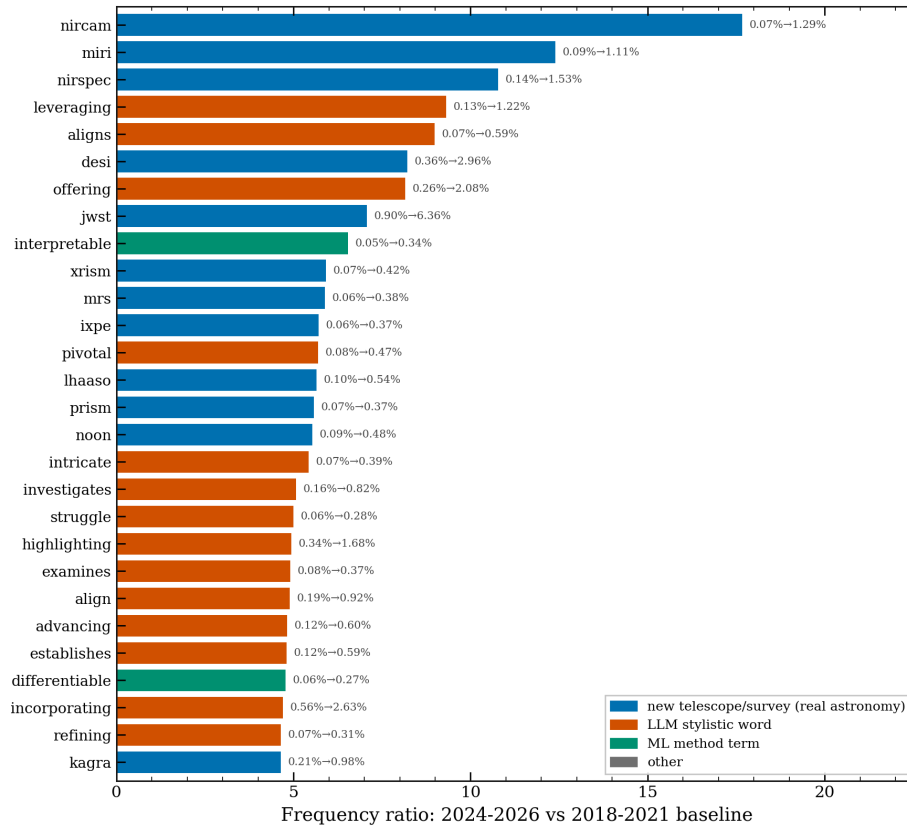


Figure 3. Words with the largest frequency increase in astro-ph abstracts, 2024–2026 versus 2018–2021. Blue: new instruments, surveys, and object classes. Red: style words. Green: technical machine-learning terms.

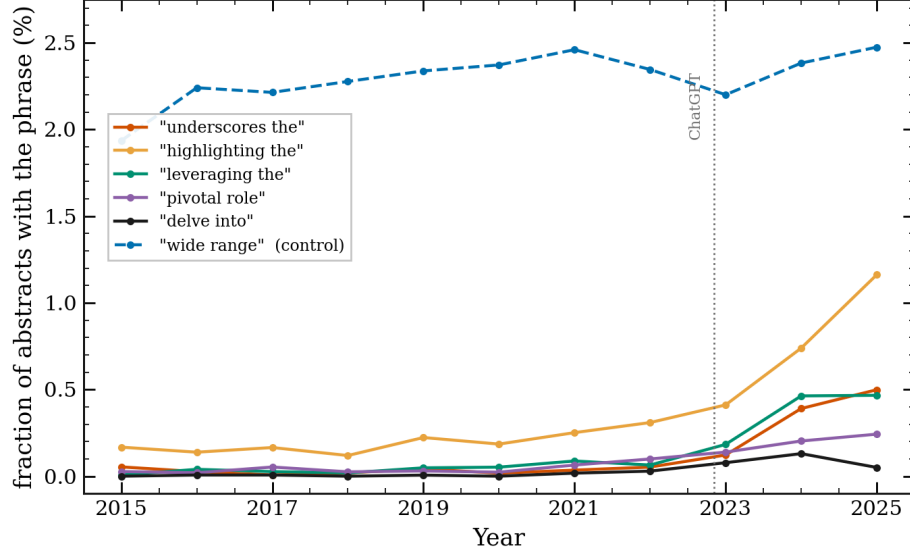


Figure 4. Fraction of abstracts containing selected two-word phrases, by year. Model-favored phrases rise after 2022; a control phrase (**wide range**) stays flat.

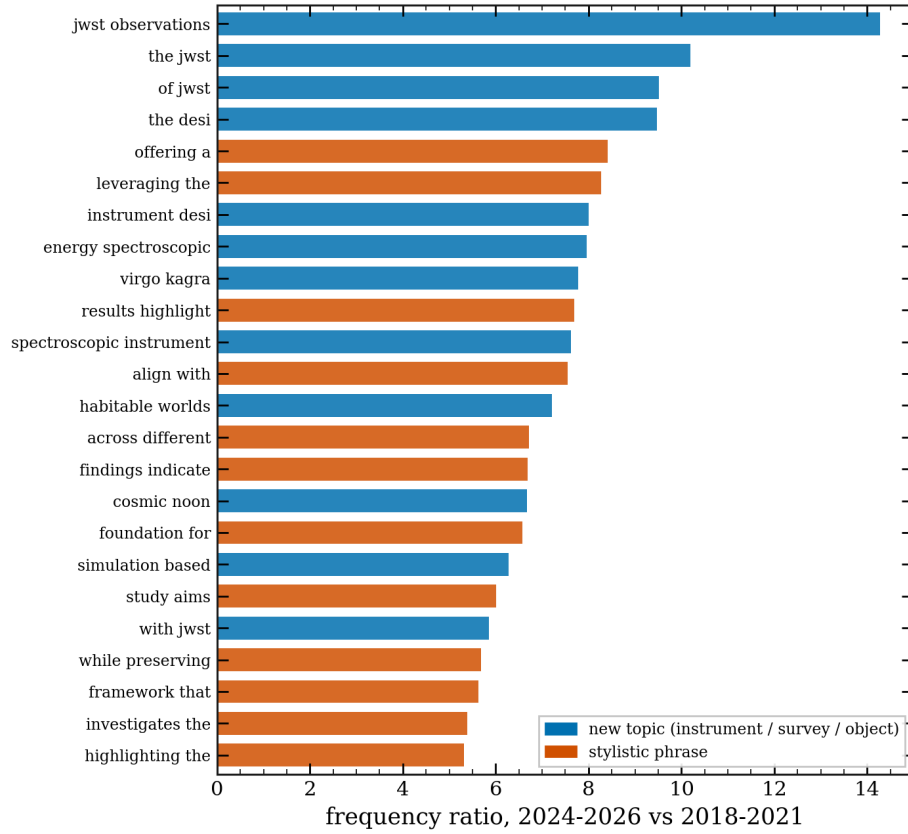


Figure 5. Two-word phrases with the largest frequency increase, 2024–2026 versus 2018–2021, split into new-topic phrases (blue) and style phrases (red).

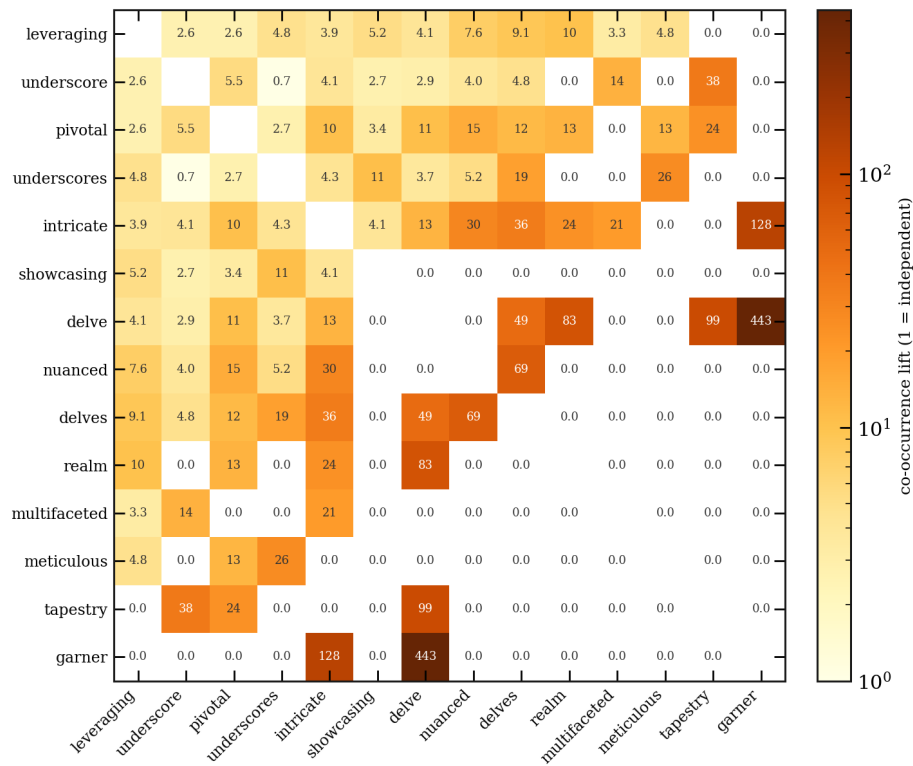


Figure 6. Co-occurrence lift for pairs of marker words in 2024–2026 abstracts. A value of one means the two words appear together as often as chance predicts. Marker words appear together far more often than that.

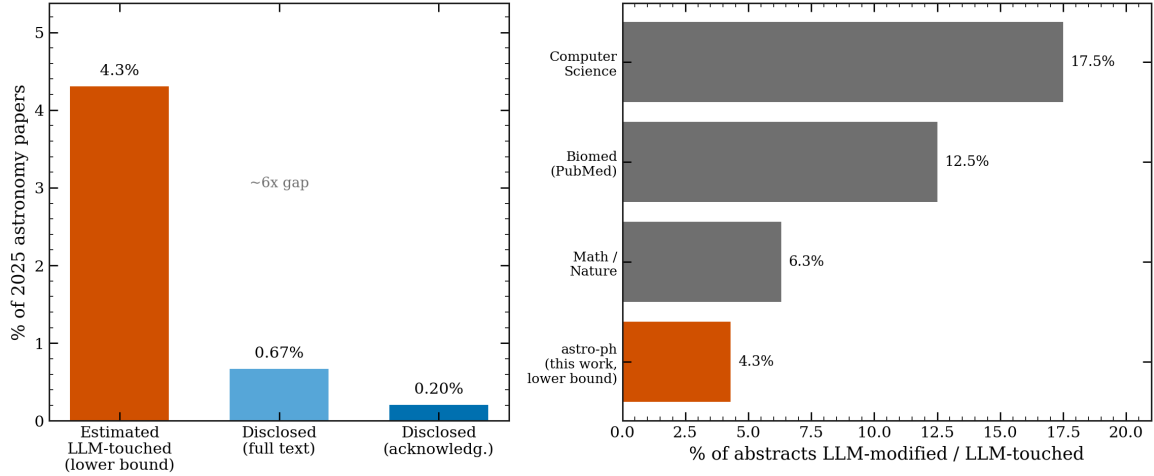


Figure 7. Left: for 2025 astronomy papers, the estimated lower bound on the model-touched fraction (about 4.3%) next to the fraction that state model use. Right: the astronomy lower bound against published estimates for other fields (Liang et al. 2024; Kobak et al. 2025).

6. A LOWER BOUND ON THE TOUCHED FRACTION

To turn the signal into a fraction we count the share of abstracts that contain at least one word from a small basket of strong, low-baseline markers, and subtract the pre-2022 rate. Let f_t be the fraction of abstracts in period t that contain at least one basket word and let f_0 be the same fraction in the 2018–2021 baseline. Our estimate is

$$\alpha_{lb} = f_t - f_0. \quad (2)$$

This is a lower bound for two reasons: a model-edited abstract need not keep any word from a short list, so f_t misses some model-touched abstracts; and by subtracting f_0 we discard any human writer who uses a basket word for ordinary reasons. Both push the estimate down. For 2025 we find $f_t = 5.8\%$ and $f_0 = 1.5\%$, so $\alpha_{lb} = 4.3\%$, and the true value is higher.

The basket is chosen for low baseline and high specificity rather than for size; broader baskets give larger numbers but add words with mixed meanings. As a check on the baseline, the control words of Section 3 give α_{lb} consistent with zero across the whole period, and moving the baseline window by a year changes α_{lb} by less than half a percentage point.

Figure 7 places this number against published estimates for other fields. Astronomy sits below computer science and biomedicine and in the same range as physics and mathematics. Because our number is a strict lower bound while the others use fuller estimators, the honest reading is that astronomy is in the same range as the rest of science, at a few percent to low double digits.

7. USING A MODEL AND SAYING SO

Authors who use a model can state so in the acknowledgments. We searched astronomy papers in the NASA Astrophysics Data System for clear terms (ChatGPT, GPT-4, “large language model”, GitHub Copilot), and left out terms that are ambiguous in astronomy, such as “Gemini” (the observatory) and “Claude” (a personal name). The count is exactly zero before 2023, which is a good check that the search is clean, and it rises every year after (Figure 8). As a share of all astronomy papers it stays below 1%: about 0.20% of 2025 papers state model use in the acknowledgments and about 0.67% mention a model anywhere in the text. Set against the 4% floor from the word counting, model use is common and stated use is rare.

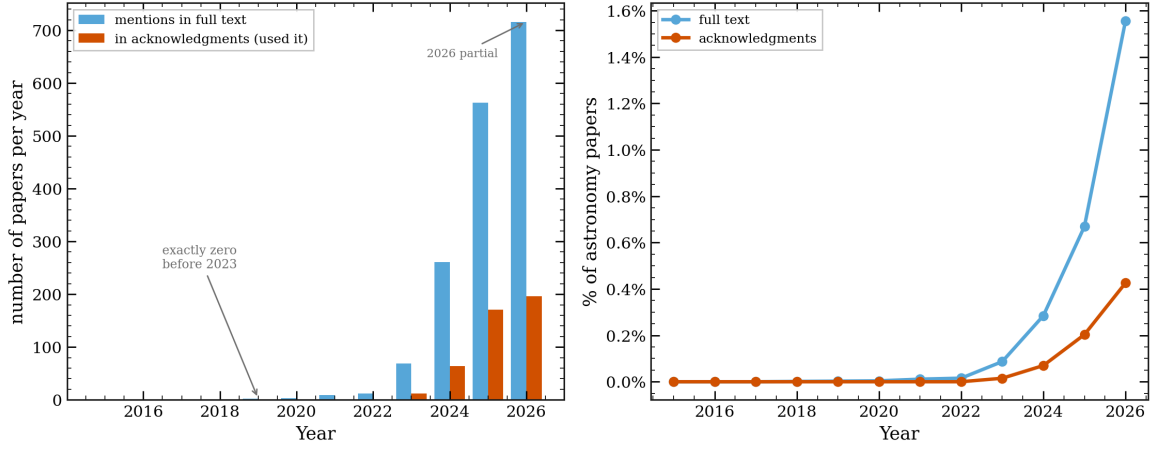


Figure 8. Left: number of astronomy papers per year that mention a model in full text (blue) or in the acknowledgments (red); the count is zero before 2023. Right: the same as a share of all astronomy papers, which stays below 1%.

8. A LADDER OF WAYS TO CATCH IT

arXiv posts the LaTeX source of most papers. A model’s boilerplate is sometimes pasted into that source and left in a comment that never reaches the PDF, such as “as an AI language model”. We downloaded the source of 3,950 papers and searched for such text. The rate is about 0.13%, and two of the few hits are papers that study models as a research topic rather than writing slips; the pre-2023 rate is exactly zero.

Placing the four lenses side by side (Figure 9) shows a range of two orders of magnitude on the same corpus. Crude source leaks (0.13%) and stated disclosure (0.19%) find the least; the word signal (4.3%, a lower bound) finds far more. A reader who trusted only the crude checks would conclude that astronomy has almost no model use, which is wrong. The word counting is the only method here that sees most of the use.

9. CITATION INTEGRITY

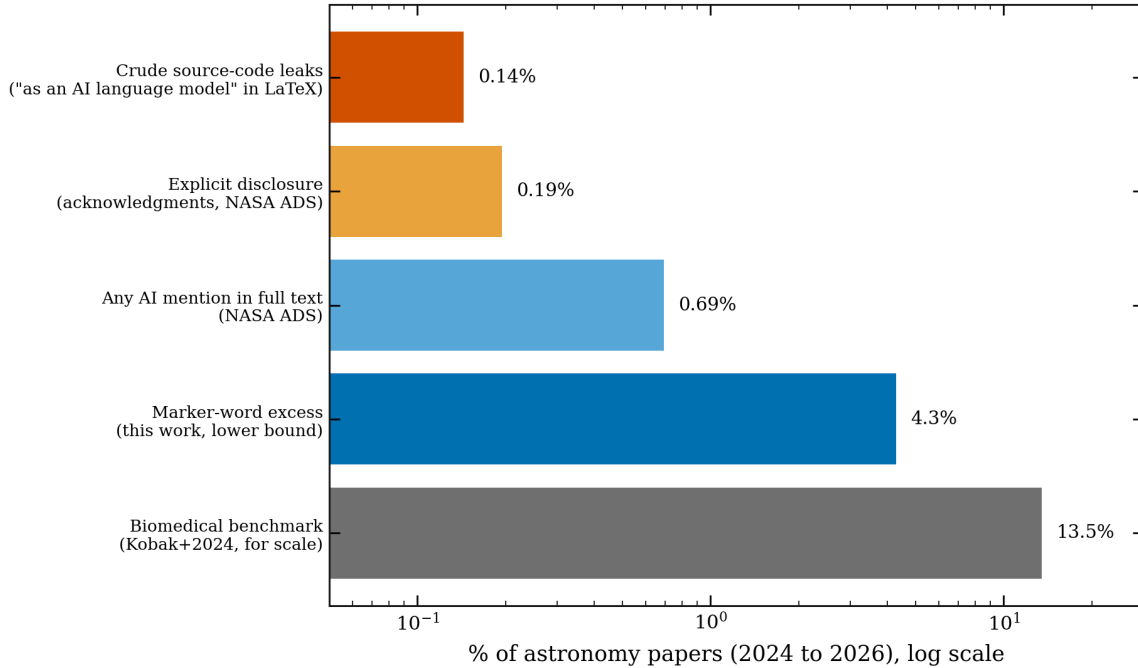


Figure 9. Four ways of looking for model use in the same 2024–2026 astronomy papers, on a log scale. Source leaks and stated disclosure find the least; the word signal finds most. The biomedical estimate (Kobak et al. 2025) is shown for scale.

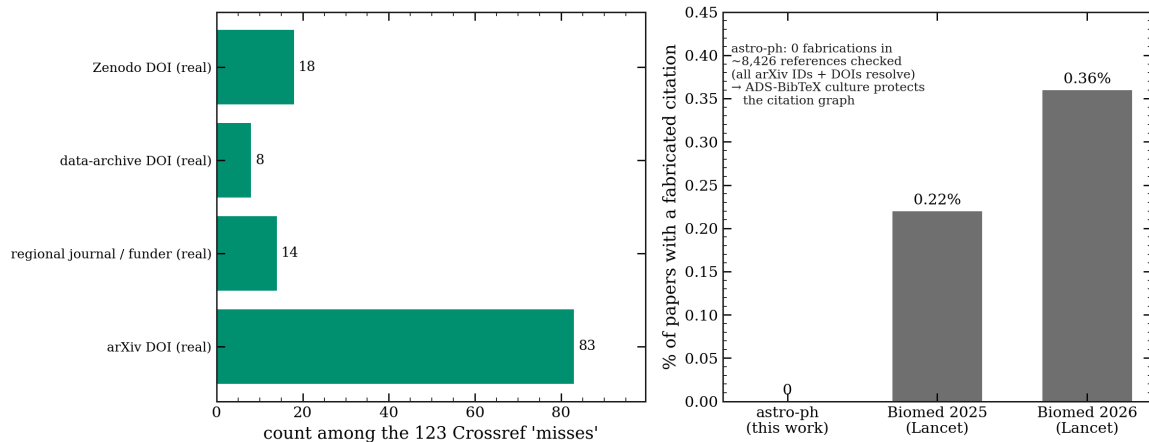


Figure 10. Left: the 123 DOIs a first check could not resolve are all real records in non-Crossref registries, not fabrications. Right: the fabricated-citation rate in astronomy (this work) against recent biomedical figures (Topaz et al. 2026).

Models sometimes produce citations to papers that do not exist, and the rate of such citations is rising in biomedicine, from about one paper in 2828 in 2023 to one in 458 in 2025 (Topaz et al. 2026). Astronomy has a defense: nearly every author builds reference lists from the NASA Astrophysics Data System, so the identifiers are real by construction. We tested this by pulling about 6,460 cited arXiv identifiers and 2,500 cited digital object identifiers (DOIs) from the source of recent papers and checking whether each one resolves. Every arXiv identifier resolves. Of the DOIs, the 123 that a first check flagged as missing are all real records held in registries that the checking service does not index (arXiv’s own system, the Zenodo archive, data archives, and regional journals), not fabrications (Figure 10). We find no fabricated citations. The reference habit of the field appears to protect its citation record against a problem that is growing elsewhere.

10. WHO, AND WHERE

The marker rate is not uniform. By subfield (Figure 11), the Instrumentation and Methods category, which is closest to software and machine learning, has the highest rate throughout, and high-energy astrophysics the lowest. The “computer science first” pattern seen across science also holds inside astronomy.

By author country (Figure 12), obtained from the affiliation field, papers from countries where English is not a first language tend toward higher marker rates, with Iran, China, and India at the high end and Japan and Russia lower. This is consistent with a reading of these tools as a writing aid for non-native English speakers, and it is a reason to treat a marker word as evidence of editing rather than of anything worse. Papers from China combine a high marker rate with one of the lowest disclosure rates, the widest gap in the sample. The affiliation search counts every country on a paper, so large collaborations appear more than once, and the small-sample countries are noisy; these numbers are first-order.

11. LOUDER, NOT MORE CAREFUL

Word counts can measure tone as well as presence. We tracked promotional words (**unprecedented**, **remarkable**, **striking**) and hedging words (**may**, **might**, **suggest**) that mark caution. The promotional set rose by about 19% over the period and follows the marker rate closely; the hedging set did not move (Figure 13). Abstracts got louder without getting more careful. This is a correlation, and competition for attention pushes promotion up on its own, but the flat caution alongside rising volume is worth recording.

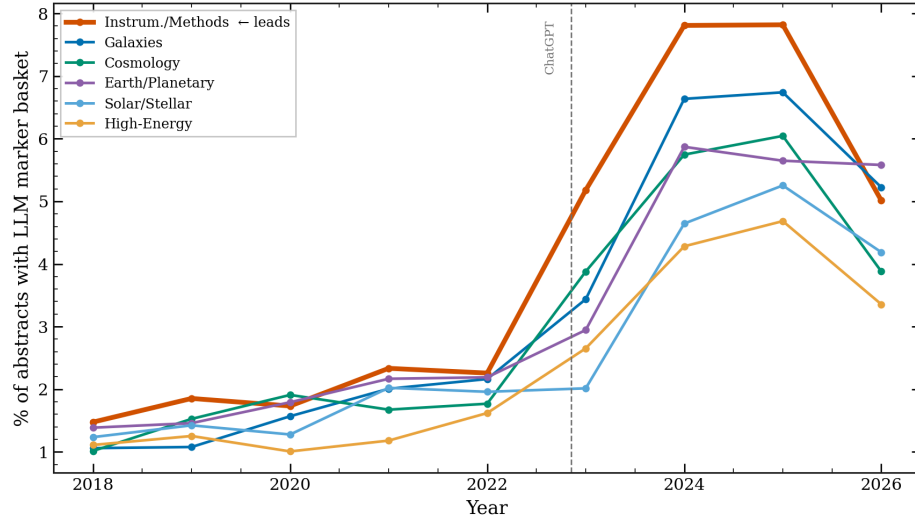


Figure 11. Marker rate by primary astro-ph subfield. Instrumentation and Methods leads; high-energy astrophysics trails.

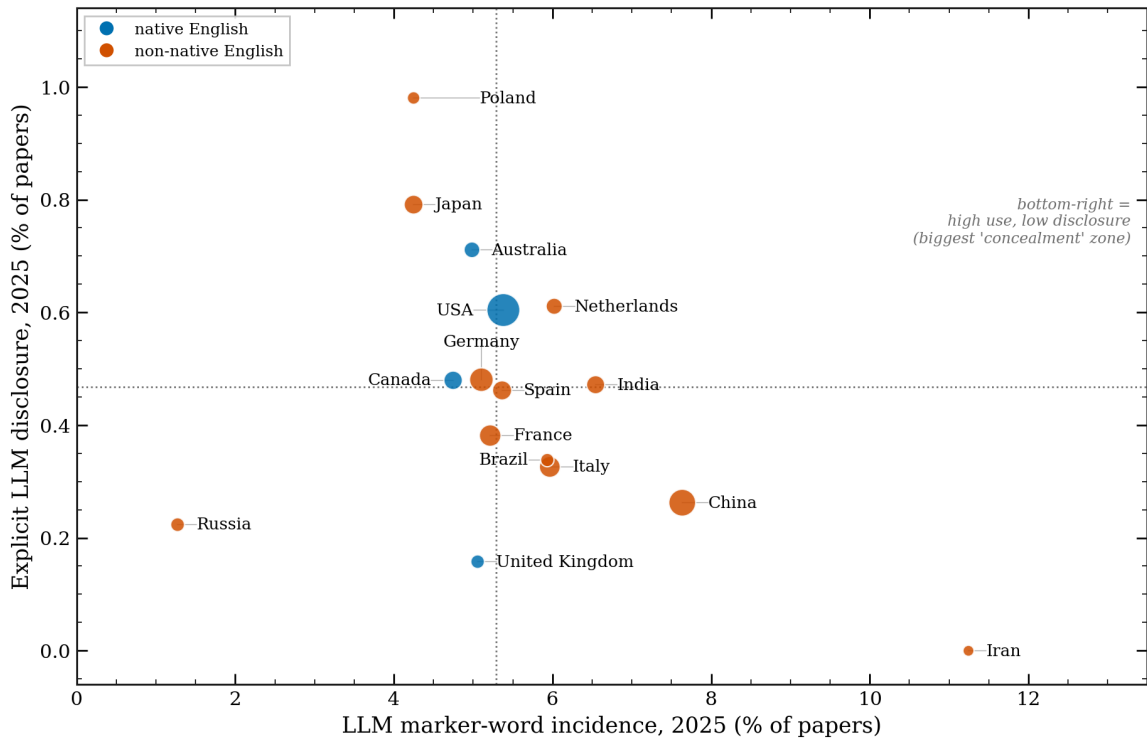


Figure 12. Marker rate against disclosure rate by author country in 2025. Blue points mark countries where English is a first language. Bubble size is the number of papers.

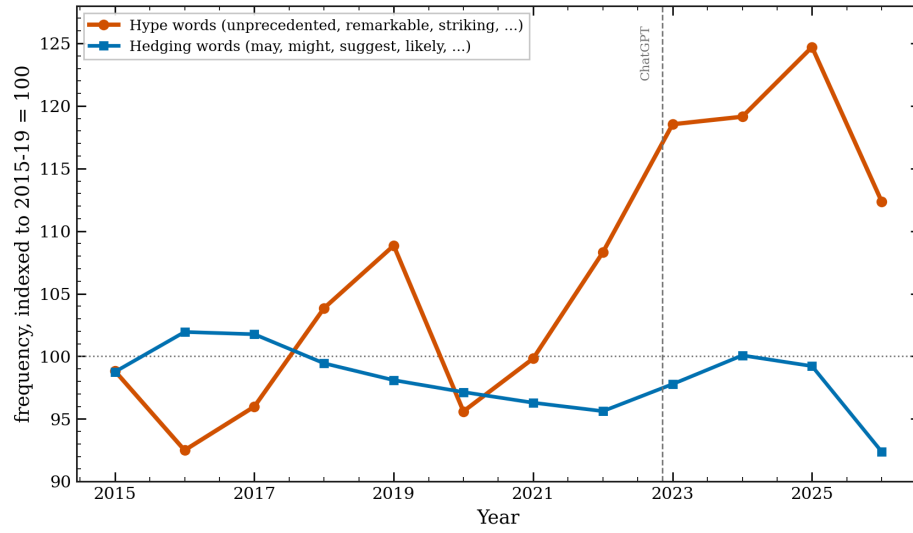


Figure 13. Promotional and hedging word frequency, indexed to the 2015–2019 mean. Promotional words rise with model use; hedging words stay flat.

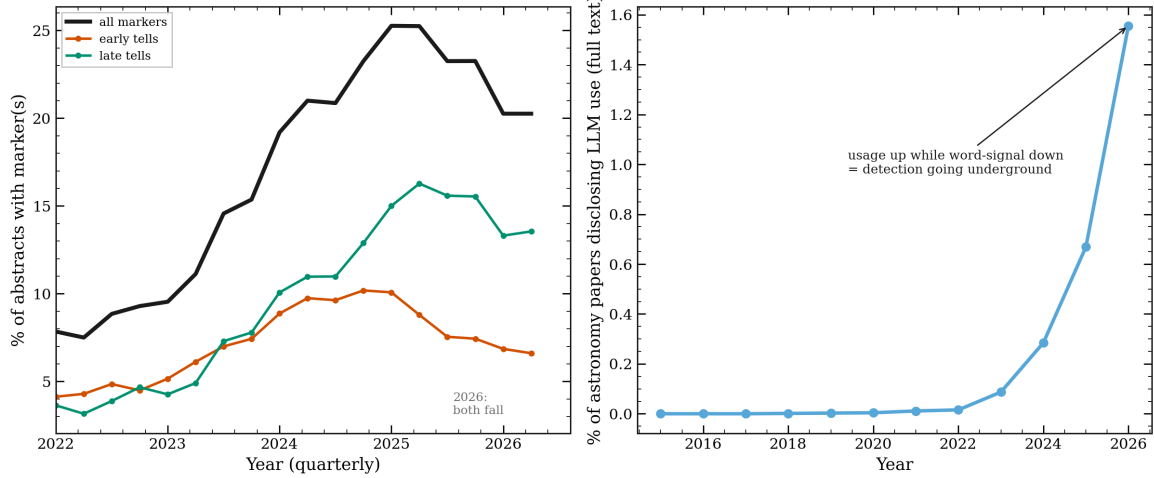


Figure 14. Left: the marker words weaken across the board in 2026, both the early tells that peaked first (**delve**, **intricate**, **pivotal**) and the later ones (**underscore**, **leveraging**). Right: over the same months, stated model use keeps rising. Use is not falling; the word signal is.

12. THE SIGNAL IS WEAKENING

The marker rate dips a little in 2026. This is not a drop in model use. Figure 14 places the marker rate next to the disclosure count over the same months: the marker words soften across the board while stated model use keeps rising, already past the full 2025 count in half a year. What weakens is the word signal, not the use. Newer models write with fewer of the known words, and authors edit the famous ones out. Any word-based number is therefore a lower bound, and the bound will slide down even as use grows.

13. DISCUSSION

The measurements above give a consistent picture. A few percent of recent astro-ph abstracts, at least, carry model-edited language; the true fraction is higher and is in the range measured for other physical sciences. Stated use is rare, below 1%, so the gap between use and disclosure is large. The marker words rise together and cluster within abstracts, which points to a single cause rather than unrelated drift. Crude “gotcha” checks find almost nothing, so the population statistics are the only reliable lens. Astronomy avoids the fabricated-citation problem seen elsewhere, most likely because of its shared reference database.

Several limits apply. First, a marker word measures model-edited language, not model-generated science; an abstract with **delve** in it is not fraudulent. Second, for many authors, and in particular for those writing in a second language, these tools lower a real barrier, so a marker word is better read as a sign of editing than of misconduct. Third, the word method has a short shelf life: as Section 12 shows, the known words fade as they become known, so a fixed list undercounts by more each year. Fourth, we read abstracts, not full papers, and abstracts are the most edited part of a paper. Finally, three uses that are easy to run together should be kept apart: a model as a copy editor, a model that drafts content, and a paper produced whole. The word counting mostly sees the first.

These results also point to method. Word lists will keep losing power, so the more durable signals are the ones tied to content rather than style: the co-occurrence structure of Section 5, which is harder to remove by editing single words, and integrity checks such as the citation audit of Section 9, which astronomy can run more cleanly than most fields because of the NASA Astrophysics Data System.

The gap between use and disclosure has a bearing on refereeing. If a few percent of abstracts are model-edited while under one percent say so, then a referee or an editor cannot assume that a clean, fluent text was written without help, and a screening tool that flags marker words will misjudge honest authors far more often than it catches misuse, because the base rate of misuse is low and the base rate of editing is not. The same tool applied to referee reports would face the same problem. A disclosure norm, stated once at submission and kept simple, would remove the guesswork at low cost, and would separate the ordinary use of a model as an editor from the uses that raise real concern.

There is also a feedback loop to watch. As a growing share of the published corpus is model-edited, the next generation of models, and any astronomy-specific models trained on astro-ph text, will learn from text that already

carries the style. Word-frequency screening will then measure the training data as much as the writing, and its output will drift. The signals that do not depend on a fixed word list, such as the co-occurrence structure and the citation checks, are the ones that will still mean what they say.

14. CONCLUSIONS

We measured the footprint of large language models in the astronomy literature from 200,547 astro-ph abstracts. The main results are:

1. The fraction of abstracts with a marker word rose from about 1.5% before 2022 to about 5.8% in 2025, while neutral control words stayed flat.
2. A conservative lower bound puts the model-touched fraction of 2025 abstracts near 4%; the true value is higher and is in the range measured for physics and mathematics.
3. Without an imported list, the data recover the same style words and phrases next to the genuinely new instrument and survey names, and the marker words occur together far more often than chance.
4. Stated disclosure stays below 1%, and crude source-code checks find near-zero pasted text; the word signal sees roughly two orders of magnitude more than either.
5. We find no fabricated citations, in contrast with a rising rate in biomedicine; the astronomy reference habit appears to protect the citation record.
6. The word signal weakens in 2026 even as disclosure rises, so word counts are a lower bound that will get harder to measure.

Whether astronomy should agree on a disclosure norm now, before a single case forces one, is a question the field can still answer on its own terms.

ACKNOWLEDGMENTS

We thank Yuan-Sen Ting for suggesting this work be written up and for pointing to the co-occurrence measurement. This research used the arXiv API and the NASA Astrophysics Data System. All data and code are available at github.com/seratsaad/llm-in-astro-ph.

Software: NumPy, pandas, Matplotlib, PyArrow

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